

Digitalization and Entrepreneurial Engagement in Developing Economies Based on Evidence from Pakistan

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Abstract— *This study examines how digitalization, measured through mobile access, computer access, and internet connectivity, affects entrepreneurship in Pakistan's non-agricultural sector. Using nationally representative data from the Pakistan Social and Living Standards Measurement (PSLM) Survey 2020–2021 and a sample of 192,670 individuals, a binary logistic regression model estimates the likelihood of engaging in entrepreneurial activity. The results show that mobile access has a strong positive association with entrepreneurship, while internet connectivity has a smaller but significant effect. In contrast, computer access is negatively associated with entrepreneurship, indicating its stronger relevance to formal employment or education. Gender and regional analyses reveal that mobile and internet access benefit rural residents and men more than women, reflecting persistent barriers to women's digital participation. Overall, the findings highlight the need to expand mobile and internet services to promote inclusive and equitable entrepreneurial growth in Pakistan and similar developing economies.*

Keywords—digital access, entrepreneurship, mobile technology, internet access, gender.

I. INTRODUCTION

The digital revolution has emerged as one of the most transformative forces reshaping contemporary economic systems, with profound implications for entrepreneurship across the globe. While this study focuses on Pakistan, it positions the country as a representative case of digitalization challenges faced by developing economies. Many countries in South Asia, sub-Saharan Africa, and Latin America share similar structural constraints, such as uneven infrastructure, low digital literacy, and gender gaps in technology use. Therefore, the insights from Pakistan can offer broader lessons for promoting digital inclusion and entrepreneurship in comparable developing contexts [1]. Digital technologies, especially mobile phones, internet connectivity, and computing devices, are increasingly recognized as critical enablers of entrepreneurial activity [2]. They facilitate access to information, reduce operational costs, enhance customer outreach, and enable participation in digital markets. In developing economies like Pakistan, where traditional infrastructure is often underdeveloped and formal employment opportunities are limited, digitalization offers a promising pathway for inclusive economic participation and grassroots entrepreneurial development [3]. Entrepreneurship, particularly in the informal and non-agricultural sectors [4],

has become an essential strategy for income generation in the face of stagnant job creation in the formal sector.

At the same time, Pakistan is experiencing a steady rise in digital adoption. [1], driven by mobile phone penetration, expanding broadband coverage, and the proliferation of digital platforms. However, despite these advancements, significant disparities remain in who benefits from digital tools and in how these tools translate into entrepreneurial outcomes. [5]. Gender, regional location, socio-economic status, and digital literacy levels continue to mediate the accessibility and impact of digital technologies on economic behavior. While policy discourse in Pakistan increasingly promotes digital entrepreneurship as part of broader economic modernization efforts [6], empirical research examining the real-world impacts of digital access on entrepreneurial engagement remains sparse. Most existing studies rely on qualitative case analyses, sector-specific evidence, or urban-centric data [7]. There is limited large-scale, quantitative research that rigorously investigates the relationship between digital access and entrepreneurship in Pakistan, especially within the informal and non-agricultural segments of the economy.

This research, to some extent, contributes to the resolution of an empirical gap by utilizing the Pakistan Social and Living Standards Measurement (PSLM) Survey 2020–2021. Within the scope of logistic regression, the research determines the effects of mobile phones, computers, and the internet on entrepreneurial involvement. The research also identifies the effect of the said variables by differentiating mobile phones and the internet. There is a possibility that the mobile phones have a higher value for entrepreneurial ventures in the remote areas as opposed to the possible high value devoid of internet, where low to no infrastructural requirements exist. Further, the mobile phones and internet have no economic value, and are used predominantly by women. Positioning the digital tools of informal economy entrepreneurship in the nexus between economic digitalization and informality promotes the entrepreneurship theory. The theory also accommodates the use of different digital devices for different entrepreneurial activities. However, it also identifies gaps in the equitable distribution of entrepreneurial avenues in Pakistan. Thus, the findings also provide rich socio/ economic instruments for the development of Pakistan.

The findings indicate that the entrepreneurial avenues are still limited for women and the remote areas. The research

strengthens the impact of digital entrepreneurial activities in Pakistan by introducing a large dataset. Further, it also advocates for the regional and gender digital gap. Third, it offers relevant digitalization policy insights regarding inclusivity and equity. More broadly, the study improves understanding of the intersections between entrepreneurial empowerment and digital technologies, as well as the particular structural changes needed to realize sustainable and inclusive entrepreneurial empowerment.

II. LITERATURE REVIEW

The link between digitalization and entrepreneurship has received increasing attention in development studies, economic sociology, and innovation research. Scholars argue that digital technologies, by lowering information asymmetries, enabling financial inclusion, and improving access to supply chains, play a catalytic role in the entrepreneurial ecosystem [8, 9]. In low- and middle-income countries (LMICs), digital tools offer unique advantages by mitigating traditional constraints such as inadequate infrastructure, limited market access, and bureaucratic inefficiencies [10]. Mobile phones, in particular, have been highlighted as the most democratizing digital tool in resource-constrained settings. Studies across sub-Saharan Africa and South Asia have demonstrated that mobile access is positively associated with increased business creation, market expansion, and improved income among small-scale entrepreneurs [11]. Mobile money platforms like M-Pesa in Kenya have revolutionized micro-entrepreneurship by providing easy, secure, and real-time financial transactions [12]. These studies support the notion that mobile technology reduces transaction costs, facilitates networking, and opens new income-generating opportunities.

The influence of different digital tools, especially computers and internet services, is more complicated. Having a laptop is more closely linked to tertiary education, formal jobs, and administrative and secretarial services than to entrepreneurship, especially in lower- and middle-income countries, where computers are mainly used in education or for business purposes [13]. This helps explain some studies that find no association, or a negative association, between access to a computer and entrepreneurship. Evidence from different countries also demonstrates the varied impacts of digital tools. In India, the use of mobile phones has enabled women in micro-entrepreneurship using WhatsApp for commerce and digital payments [14].

In Bangladesh, the combination of Grameenphone and BRAC microfinance services enhanced women's business activities in rural areas [15]. Also, in Kenya, M-Pesa was a significant boost to the growth of small and medium-sized enterprises by lowering transaction costs and increasing access to money [12]. On the other hand, in Brazil and Peru, access to the internet and computers has mainly supported young people to set up digital businesses and online enterprises [16]. These studies demonstrate that although access to digital technologies is beneficial for entrepreneurship, its impact varies by social status, gender, and the availability of physical infrastructure. Building on this information, the current research seeks to identify the equivalents of such technology-specific patterns in Pakistan.

Internet access is widely regarded as an enabler of innovation and market expansion by allowing entrepreneurs to reach broader customer bases, engage in e-commerce, and access business information. However, its benefits depend on affordability, connectivity quality, and user capabilities, and without complementary skills, digital connectivity may reinforce existing inequalities [17]. In Pakistan, limited digital skills and high internet costs constrain the entrepreneurial potential of internet access, particularly for disadvantaged groups [18]. Gender further mediates this relationship: globally, women are less likely than men to own digital devices or use the internet [19], a gap reinforced in Pakistan by socio-cultural norms, mobility restrictions, and domestic roles [20, 21]. Even where access exists, productive use is not guaranteed, underscoring the need for institutional support, capacity building, and cultural change to enable women to benefit from digitalization [22] fully. Acting at a regional level, disparities shape potential Colombian digital entrepreneurs. urban Colombian entrepreneurs due to disparities in infrastructure, networks, and greater exposure to and use of digital tools, relative to rural Colombian entrepreneurs, who experience shortages of power and broadband, weak institutional support, and lower levels of digital finance trust. The rural digital divide, while partly mitigated, reinforces the need for policies that include digital tools to provide a starting point for addressing systemic inequities. The literature also shows the existence of supportive environments. such as legal and regulatory frameworks, without which digital access does not result in changes to entrepreneurial outcomes. In a context like Pakistan, where institutional support is weak and informal entrepreneurship is rampant, digital devices will not cross the existing structural divide. In summary, the literature reveals that digitalization has the potential to enhance entrepreneurship, but its effects are heterogeneous and context dependent. Access to mobile phones and internet services appears to be more strongly correlated with entrepreneurship than access to computers. Yet these benefits are often mediated by gender, location, and socio-economic status. This underscores the importance of not only expanding access but also designing targeted interventions that address deeper structural and cultural constraints. The current study contributes to this literature by conducting a large-scale empirical investigation of how digital access—measured by mobile, PC, and internet availability—affects entrepreneurial engagement in Pakistan, with specific attention to gender and regional differences.

TABLE I. COMPARATIVE EVIDENCE ON DIGITALIZATION AND ENTREPRENEURSHIP IN DEVELOPING ECONOMIES

Country	Dataset / Study	Main Findings on Digitalization & Entrepreneurship
Pakistan	PSLM 2020–21	Mobile (+), Internet (+), Computer (–)
India	NSSO / World Bank studies [26]	Mobile phones increased female micro-entrepreneurship
Bangladesh	BRAC & Grameenphone studies [27]	Mobile + microfinance increased rural women's business participation
Kenya	M-Pesa [12]	Mobile money expanded SMEs and household businesses
Brazil	National ICT Survey [28]	PC + Internet-supported youth-led digital start-ups
Peru	ICT and Youth Employment Survey [28, 29]	Internet access is associated with digital entrepreneurship among youth

III. ECONOMETRIC MODEL

The logit model is particularly well-suited for this study, as the dependent variable, entrepreneurship status, is binary in nature. Specifically, individuals are classified as either entrepreneurs (i.e., employers or self-employed in non-agricultural sectors) or non-entrepreneurs [30]. In such contexts, ordinary least squares (OLS) regression is inappropriate due to the violation of assumptions regarding linearity, normality, and homoscedasticity [31]. The logistic regression model overcomes these limitations by estimating the probability of entrepreneurial engagement through a logistic transformation of the linear function [32], ensuring that predicted values remain bounded between 0 and 1. Furthermore, the model facilitates interpretation in terms of odds ratios, which are especially useful for understanding how access to digital tools, such as mobile phones, computers, and the internet, affects the likelihood of entrepreneurship. By incorporating individual-level covariates and regional controls, the logit model provides a robust framework for isolating the unique contribution of digital access to entrepreneurial outcomes in a multivariate setting. The econometric model is as follows:

$$\log\left(\frac{P(\text{Entrepreneur}=1)}{1-P(\text{Entrepreneur}=1)}\right) = \beta_0 + \beta_1 \text{Mobile} + \beta_2 \text{Computer} + \beta_3 \text{Internet} + \beta_4 \text{Province} + \beta_5 \text{Region} + \beta_6 \text{Gender} + \beta_7 \text{Age} + \beta_8 \text{Education} + \beta_9 \text{Income} + \epsilon$$

TABLE II. DEFINITION OF VARIABLES

Variable Name	Description	Type	Coding Values /
Entrepreneur	Indicates whether the individual is engaged in entrepreneurial activity as an employer (small or large) or self-employed in a non-agricultural sector	Dependent Variable (Binary)	1 = Employer (<10 persons), Employer (≥10 persons), or Self-employed non-agri 0 = Otherwise
Mobile Access	Indicates whether the individual owns or has access to a mobile phone	Key Variable (Binary)	1 = Yes 0 = No
Computer Access	Indicates whether the individual owns or has access to a desktop/laptop/tablet	Key Variable (Binary)	1 = Yes 0 = No
Internet Connectivity	Indicates whether the individual has access to the internet	Key Variable (Binary)	1 = Yes 0 = No
Province	Province of residence	Categorical	1 = Khyber Pakhtunkhwa 2 = Punjab 3 = Sindh 4 = Balochistan
Region	Area of residence	Categorical	1 = Rural 2 = Urban
Gender	Biological sex of the individual	Binary	1 = Male 2 = Female
Age	Age of the individual in completed years	Continuous	In years (e.g., 18, 35, 60)
Education	Highest level of formal education completed	Continuous	0 = No education 21 = PhD (scale from 0 to 21)
Household Income	Total income of the household from both labor and non-labor sources	Continuous	Sum of all household earnings (in local currency)

In this specification, each coefficient is interpreted as a change in the log-odds of being an entrepreneur associated with a one-unit change in the corresponding explanatory variable, holding all other variables constant. The odds ratios (i.e., exponentiated coefficients) are reported for easier interpretation. The definition of the variables used in the study is presented in Table I. A few limitations should be noted when interpreting the results. The analysis relies on self-reported measures of digital access, which may not accurately reflect actual usage or digital skill levels. Individuals might overstate or understate their engagement with technology, introducing potential reporting bias. Moreover, the cross-sectional nature of the PSLM data limits the ability to establish causal relationships, as the direction of influence between digital access and entrepreneurship cannot be fully verified. Unobserved factors, such as personality traits or local institutional quality, may also affect entrepreneurial outcomes. Despite these constraints, the large and nationally representative sample enhances confidence in the robustness of the findings and the reliability of the observed associations.

IV. DATA AND DESCRIPTIVE ANALYSIS

A. Data source

For the year 2020-2021, the Pakistan Bureau of Statistics (PBS) has made available the Social and Living Standards Measurement (PSLM) Survey that this research relies upon. The PSLM covers a wide array of aspects regarding households on a national scale, including arete, education, health, income, and access to minimal services, along with a few others. For PSLM data, only data for the non-agricultural sector were analyzed while excluding entries with no information on the survey. The finalized sample includes 192,670 respondents with non-agricultural economic activities, out of which a sample of over 190,000 is available, which will enable remote and ample research on the nexus of digital access and entrepreneurship.

B. Descriptive statistics

Table II provides summary statistics for the main variables under consideration. The dependent variable Entrepreneur is a dummy variable, which takes the value of one for those who are non-agricultural employees, employers, or self-employed. The average is 0.262, which means that 26.2 % of the observations are participating in entrepreneurial activities, while 73.8 % are non-entrepreneurs. This confirms the predominance of wage employment in the non-agricultural sector of the economy of Pakistan. The standard deviation of the entrepreneurial status (0.44) is a proxy for the moderate range of educational dispersion, which makes a case for this study. In particular, the role of factors such as digital accessibility.

The descriptive statistics show considerable disparities in digital leave. More than three-quarters (77.1%) of the population have a mobile device, with a small minority (22.9%) not having access to this readily available digital resource. However, the digital divide remains considerable, with only a small portion of the population having access (less than 25%) to advanced ICTs. Internet access is also very limited among the sample, with 25.1% having internet access, while 74.9% are isolated from the digital economy (online trading, digital marketing, Internet banking). The limited access to computers as well as the Internet leaves most of the

population in a digital economy, in spite of high mobile penetration, restricting the overall digital participation of the economy.

Table II demonstrates the geographic and demographic statistics of the sample surveyed. In Punjab, the respondents are most concentrated at 50.8% of the sample, while Sindh, Khyber Pakhtunkhwa, and Balochistan are at 22.6% 17.2%, and 9.3%, respectively, which is consistent with the population percentages in the main administrative divisions of the country. In terms of geography, 66% of the respondents are in the rural and 34% in the urban regions of the country, which suggests the existence of rural-urban inequalities in digital accommodations and in economically entrepreneurial opportunities. The sample is quite skewed, with 84.9% male and 15.1% female, which is either the result of the employment-driven nature of the questionnaire or the employment structure of the non-agricultural sectors of the economy. All of these reasons are the basis of why the differences of the regions, the provinces, and the demographics of the respondents have been controlled.

TABLE III. DESCRIPTIVE STATISTICS

Variable	Obs	Mean	Std. Dev.	Min	Max
Entrepreneur					
No	192670	.738	.44	0	1
Yes	192670	.262	.44	0	1
Mobile use					
No	192670	.229	.42	0	1
Yes	192670	.771	.42	0	1
PC use					
No	192670	.912	.283	0	1
Yes	192670	.088	.283	0	1
Internet use					
No	192670	.749	.433	0	1
Yes	192670	.251	.433	0	1
Province					
Khyber Pakhtunkhwa	192670	.172	.378	0	1
Punjab	192670	.508	.5	0	1
Sindh	192670	.226	.418	0	1
Balochistan	192670	.093	.291	0	1
Region					
Rural	192670	.66	.474	0	1
Urban	192670	.34	.474	0	1
Gender					
Male	192670	.849	.358	0	1
Female	192670	.151	.358	0	1
Age	192670	34.36 9	12.941	10	98
Education	192670	5.653	5.311	0	21
Log(hhincome)	192670	10.20 6	1) 1. 401	0	15.47 2

The descriptive statistics for the continuous control variables further characterize the socio-economic profile of the sample. The mean age is 34.4 years (SD = 12.94), with values ranging from 10 to 98 years, covering a broad working-age and older population relevant for analyzing age-related entrepreneurial patterns. Education, measured in years of schooling, averages 5.65 years (SD = 5.31), ranging from no formal education to 21 years, reflecting wide variation but generally low educational attainment consistent with national trends, particularly in rural and disadvantaged areas. Household income, expressed in logarithmic form to address skewness, has a mean of 10.21 (SD = 1.40), with values between 0 and 15.47, indicating substantial income dispersion. The log transformation facilitates clearer interpretation of income effects in regression analysis. Collectively, these variables capture key dimensions of human capital and economic status relevant for explaining variation in entrepreneurial engagement alongside digital access.

V. RESULTS AND DISCUSSION

A. Baseline model

Table III summarizes five logistic regression models examining how digital access influences entrepreneurship in Pakistan's non-agricultural sector. Models 1 to 3 include mobile, computer, and internet access separately, while Models 4 and 5 test their combined effects. Across all models, mobile access shows a strong positive association with entrepreneurship, computer access remains negative, and internet connectivity has a more negligible but consistent positive effect. These results indicate that different digital tools shape entrepreneurship in distinct ways. Mobile phones appear to be the most accessible and influential form of digitalization, especially where infrastructure and capital are limited. Computers, in contrast, are more linked to formal jobs and education, reflecting structural divides in Pakistan's economy. Internet access contributes modestly by improving communication and market reach, but is limited by cost and coverage.

Similar trends are observed across other developing economies, suggesting that expanding mobile and internet infrastructure, particularly in rural and low-income regions, could promote broader entrepreneurial growth. The results in Table III reveal that individuals with mobile phone access are approximately 97% more likely to be engaged in entrepreneurial activity than those without mobile access, as shown in Model 1. This interpretation is based on the odds ratio of 1.968, which translates to a 96.8% increase in the odds of being an entrepreneur. This effect remains highly consistent and statistically significant across Models 4 and 5, where mobile access continues to increase the odds of entrepreneurship by approximately 96.4% and 92.7%, respectively.

The robustness of these estimates underscores the pivotal role of mobile technology in supporting entrepreneurial engagement, particularly by enhancing access to markets, information, and communication in low-resource environments. The regression results presented in Table III show a consistently negative and statistically significant association between PC access and the likelihood of being engaged in entrepreneurship. In Model 2, individuals with access to a personal computer (desktop, laptop, or tablet) are approximately 45.4% less likely to be entrepreneurs compared to those without such access, as indicated by an odds ratio of

0.546. This pattern remains stable in Models 4 and 5, where the odds of entrepreneurship among PC users are 45.0% and 47.5% lower, respectively. These findings suggest that, unlike mobile phones or internet access, PC ownership may be more closely associated with formal wage employment, education-related activities, or administrative tasks than with self-directed entrepreneurial work. The results underscore the nuanced role that different forms of digital access play in shaping entrepreneurial engagement, particularly in contexts where informal and mobile-based enterprises dominate.

The positive relationship between Internet access and entrepreneurial activity is statistically significant, as shown in Table III. Individuals with Internet access are, on average, 12.7% more likely to be entrepreneurs (odds ratio = 1.127). This effect is strengthened when Internet access (Model 5) increases the odds of entrepreneurship by 16.0% (odds ratio = 1.160) after controlling for access to a mobile phone and a computer. This demonstrates that access to the Internet is likely to facilitate entrepreneurship through greater access to information, easier product and service marketing, greater customer acquisition, and faster transaction processing. This likely results in lower barriers to starting a business in poorer regions, where there is no physical infrastructure or formal employment.

The estimates in Table III also indicate considerable differences across provinces by region (with KPK as the chosen comparison category). Those in Sindh are the least likely to be entrepreneurs, with odds ratios between 0.578 and 0.606, indicating they are 39.4% to 42.2% less likely to engage in entrepreneurship. People in Balochistan also have reduced odds of being entrepreneurial, with odds ratios of 0.819 and 0.857, indicating they are 14.3% to 18.1% less likely to be entrepreneurial. While the results for Punjab are apathetic, as it appears from models 2 and 3, there is a very small likelihood (though probably insubstantial) of odds ratios being 0.962 and 0.971; this effect indeed becomes statistically insignificant for the complete models (4 and 5), showing no reliable difference as compared to KPK.

Overall, these trends indicate considerable regional differences that are evidently due to factors surrounding entrepreneurial activity, such as infrastructure, the region's economy, access to necessary resources for entrepreneurship, and the institutions supporting entrepreneurial activity. Individuals living in urban areas are more likely to be entrepreneurs than those living in rural areas, even in rural-specific models. All five models show significant positive results regarding urban residency and entrepreneurship, with urban residents being 41.2%- 48.4% more likely to participate in entrepreneurial activities. One possible explanation for this trend is the urban advantage phenomenon, as reflected in the results: urban areas have better, more developed infrastructure, access to funding, financial technology, and a more varied consumer market. The results reflect the case of the rural regions' gap and signal to policymakers the need for entrepreneurship and the case of rural areas with developed access.

Participation in Table III indicates a robust and statistically significant inverse relationship between entrepreneurial involvement and being female. The odds ratios for women range from 0.271 to 0.401 across all five models. Therefore,

women are 59.9% to 72.9% less likely to be entrepreneurs compared to men, *ceteris paribus*. The gender gap remains significant after accounting for digital access, region, and socioeconomic levels. This indicates substantial structural and cultural barriers to women's economic activity in Pakistan. This also underscores the importance of gendered policy frameworks to address the obstacles such as limited access to finance and markets, restrictive mobility, and socially restrictive norms against women's entrepreneurship. Age also shows a strong, statistically significant positive relationship with entrepreneurship, with odds ratios between 1.031 and 1.033 across the models. This means that, other things being equal, an additional year of age increases the odds of being an entrepreneur by about 3.1% to 3.3%. This constant effect means that, over the years, an individual's potential to be entrepreneurial increases through the accumulation of experience, skills, networks, and financial resources. This aligns with the life cycle theories of entrepreneurship. The relationship between education and entrepreneurship is slight and inconclusive. In Model 1, the odds ratio of $[0.994]$ suggests that, on average, there is a slight decrease in the odds of being in the entrepreneur sector in the future. However, this is countered in Model 2, which suggests there is a $[2\%]$ increase (odds ratio $[1.020]$).

In model 5, fully specified, the effect of education is rather small and also somewhat heterogeneous, adding only around 0.5 percentage points to the probability of one engaging in entrepreneurial activities for each year's increase in education, as represented by the odds ratio of 1.005. Education shows a statistically significant connection with entrepreneurship in most of the models. Nevertheless, the small size of the effect in virtually all of the models evaluated suggests that, by itself, formal education is very unlikely to cause a significant increase in the entrepreneurial activity of the population. This effect may possibly originate from a higher selection of the highly educated into paid employment, and non-entrepreneurial jobs in particular, especially if the position has a decent level of income and remuneration that is above the entrepreneurial profit horizon. Additionally, this effect suggests education operates in the context of the study more as an enabling or ancillary factor than as a major determinant of entry into entrepreneurship.

In contrast, household income exhibits a strong and consistently positive association with entrepreneurial activity, as reported in Table III. All five model estimations reveal a uniform pattern, with odds ratios ranging from 1.038 to 1.053, indicating that a one percent increase in household income raises the probability of entrepreneurial engagement by approximately 3.8 to 5.3 percent. This robust relationship highlights the critical role of income in facilitating entrepreneurship, as higher income levels reduce financial constraints, ease risk-taking behavior, and enhance access to credit and productive assets necessary for business initiation and expansion. Additionally, higher income levels provide a financial buffer that allows individuals to absorb potential losses during the early stages of entrepreneurial activity. Overall, the stability and strength of income effects across all model specifications underscore income as a key determinant of entrepreneurial activity in Pakistan's non-agricultural sector.

TABLE IV. ESTIMATES OF LOGISTIC REGRESSION MODEL – DIGITALIZATION AND ENTREPRENEURSHIP

Variables	(1)	(2)	(3)	(4)	(5)
	<i>Model 1</i>	<i>Model 2</i>	<i>Model 3</i>	<i>Model 4</i>	<i>Model 5</i>
Mobile	1.968*** (0.0359)			1.964* ** (0.0359)	1.927* ** (0.0355)
PC		0.546*** (0.0124)		0.550* ** (0.0125)	0.525* ** (0.0121)
Internet			1.127* ** (0.0161)		1.160* ** (0.0169)
KPK (base)					
Punjab	1.001 (0.0149)	0.962*** (0.0143)	0.971* * (0.0144)	0.993 (0.0148)	0.993 (0.0148)
Sindh	0.606*** (0.0114)	0.578*** (0.0108)	0.580* ** (0.0108)	0.602* ** (0.0113)	0.599* ** (0.0113)
Balochistan	0.857*** (0.0190)	0.819*** (0.0181)	0.833* ** (0.0184)	0.843* ** (0.0187)	0.841* ** (0.0187)
Urban	1.412*** (0.0171)	1.484*** (0.0180)	1.440* ** (0.0175)	1.437* ** (0.0175)	1.419* ** (0.0174)
Female	0.381*** (0.00873)	0.284*** (0.00602)	0.271* ** (0.00572)	0.401* ** (0.00924)	0.400* ** (0.00922)
Age	1.031*** (0.000436)	1.033*** (0.000430)	1.033* ** (0.000432)	1.032* ** (0.000437)	1.032* ** (0.000442)
Education	0.994*** (0.00111)	1.020*** (0.00125)	1.000 (0.00121)	1.009* ** (0.00126)	1.005* ** (0.00132)
Ln(hhincome)	1.040*** (0.00471)	1.053*** (0.00491)	1.038* ** (0.00472)	1.050* ** (0.00488)	1.044* ** (0.00484)
Constant	0.0519*** (0.00270)	0.0722*** (0.00371)	0.0874*** (0.00436)	0.0446*** (0.00237)	0.0466*** (0.00247)
Observations	192,670	192,670	192,670	192,670	192,670

Standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

B. Heterogeneity analysis

Table IV shows differences in the analysis comparing the impact of mobile phones, computers, and the internet relative to the total sample on entrepreneurship in rural and urban areas, and separately for men and women. The separate logistic regressions reveal some subgroup-specific features that would otherwise become concealed in the averaged estimates. The derived Figure 1 enhances the visual comparison of the odds ratios for the various subgroups in Table IV, where the interpretational example is amplified and discrepancies in digital resources and entrepreneurship participation are underscored. The evidence suggests that digital resources are more helpful for some than for others, with this pattern more evident by gender and geographical location, highlighting the value of differentiated policy responses. Analyses indicate a consistently positive and statistically significant relationship between mobile accessibility and entrepreneurial activity across all subsamples. Access to mobile telephony and the entrepreneurship relationship in the total sample (Model 1) shows that mobile accessibility increases the probability of entrepreneurship by about 93% (odds ratio = 1.927). This effect is most pronounced in the rural context, where mobile penetration increases the odds of engaging in entrepreneurial activity by 106.5% (odds ratio = 2.065), underscoring the importance of mobile telephony in rural settings where entrepreneurship is most needed. In the urban context, the effect remains positive but is lower at 43.1% (odds ratio = 1.431), even though it remains statistically significant, likely because urban areas have more alternative digital technologies. Disaggregated by gender, mobile access increases the entrepreneurial odds by 93.6% for men and 57.5% for women. These findings suggest that mobile technology enhances entrepreneurship for all, but the enhancement is more significant for rural men, and this calls for improved digital inclusion for women and informal entrepreneurs in urban areas.

The estimates in Table IV indicate a consistently negative and statistically significant association between PC access and entrepreneurial activity across all subsamples. In the full sample (Model 1), access to a personal computer reduces the odds of entrepreneurship by 47.5% (odds ratio = 0.525). This negative relationship persists across regional and gender-based subsamples, with entrepreneurial odds declining by 48.5% in rural areas and 42.6% in urban areas. Gender-disaggregated results show that PC access lowers the likelihood of entrepreneurship by 46.4% for men and by 62.7% for women (odds ratio = 0.373), indicating a

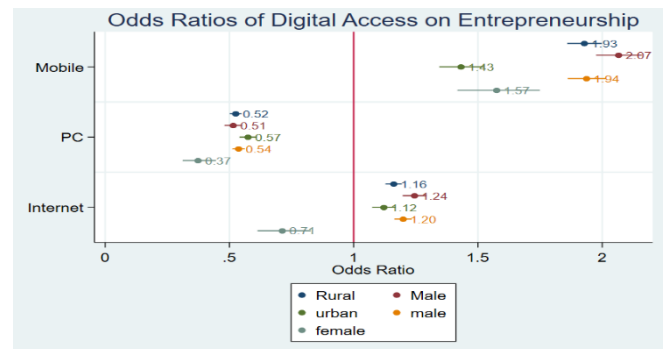


Fig. 1. Odds ratios of digital access on entrepreneurship across subgroups

substantially more substantial effect among female respondents. These patterns suggest that PC ownership is more closely associated with formal employment or educational activities than entrepreneurial engagement, particularly for women. Overall, the findings highlight that different digital technologies exert heterogeneous effects on entrepreneurship, underscoring the need for differentiated, demographically sensitive digital policy interventions.

Table V presents compelling evidence on the role of internet connectivity in shaping entrepreneurial outcomes, though the effect varies notably across population subgroups. In the overall sample, individuals with internet access are 16% more likely to be entrepreneurs (odds ratio = 1.160), highlighting the enabling role of online connectivity in accessing markets, information, and business networks.

TABLE V. Estimates of the Logistic Regression Model: Digitalization and Entrepreneurship Across Sub-Samples

VARIABLES	(1) Whole Sample	(2) Rural Sample	(3) Urban Sample	(4) Male Sample	(5) Female Sample
Mobile	1.927** *	2.065** *	1.431** *	1.936** *	1.575** *
	(0.0355)	(0.0470)	(0.0455)	(0.0391)	(0.0844)
PC	0.525** *	0.515** *	0.574** *	0.536** *	0.373** *
	(0.0121)	(0.0182)	(0.0177)	(0.0128)	(0.0340)
Internet	1.160** *	1.244** *	1.121** *	1.198** *	0.712** *
	(0.0169)	(0.0243)	(0.0242)	(0.0178)	(0.0543)
KPK					
Punjab	0.993 (0.0148)	1.005 (0.0169)	1.058* (0.0355)	0.992 (0.0153)	1.194** (0.0850)
Sindh	0.599** *	0.561** *	0.687** *	0.585** *	1.034 (0.0842)
	(0.0113)	(0.0148)	(0.0241)	(0.0114)	(0.0842)
Balochistan	0.841** *	0.840** *	0.910* (0.0437)	0.821** *	1.338** (0.136)
	(0.0187)	(0.0212)	(0.0437)	(0.0187)	(0.136)
Urban	1.419** *			1.352** *	2.895** *
	(0.0174)			(0.0171)	(0.140)
Female	0.400** *	0.339** *	0.584** *		
	(0.0092 2)	(0.0103)	(0.0209)		
Age	1.032** *	1.031** *	1.036** *	1.035** *	1.008** *
	(0.0004 42)	(0.0005 63)	(0.0007 22)	(0.0004 62)	(0.0016 6)
Education	1.005** *	1.010** *	0.994** *	1.002 *	1.041** *
	(0.0013 2)	(0.0017 1)	(0.0020 6)	(0.0013 5)	(0.0057 4)
Ln(hhincome)	1.044** *	1.081** *	0.993 (0.0067 1)	1.048** *	1.007 (0.0178 5)
	(0.0048 4)	(0.0070 4)	(0.0067 1)	(0.0050 5)	(0.0178 5)
Constant	0.0466* **	0.0317* **	0.126** *	0.0426* **	0.0417* **
	(0.0024 7)	(0.0023 1)	(0.0106)	(0.0023 7)	(0.0081 4)
Observations	192,670	127,134	65,536	163,639	29,031

Standard error in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

The positive effect is most potent in rural areas, where internet access is associated with a 24.4% increase in the odds of entrepreneurship (odds ratio = 1.244), reflecting the critical role of connectivity in overcoming geographic and infrastructural constraints. In urban areas, the effect remains

positive but more modest, at 12.1%. Among male respondents, internet access raises the odds of being an entrepreneur by 19.8%. In contrast, the effect for female respondents is adverse, with an odds ratio of 0.712, suggesting a 28.8% decrease in the likelihood of entrepreneurship among women with internet access. This gender divergence may reflect differences in how digital tools are used, or barriers that prevent women from leveraging internet connectivity for business purposes—such as digital literacy gaps, safety concerns, or unequal access to online economic opportunities. These findings point to the importance of gender- and context-sensitive digital inclusion strategies to ensure equitable entrepreneurial participation in the digital economy.

These subgroup patterns also echo global evidence on the digital gender divide and rural–urban disparities in digital access. Women in many developing economies face similar barriers to leveraging internet tools for business due to social norms, safety concerns, and skill gaps [16, 19]. Likewise, studies from Africa and South Asia show that mobile phones are especially transformative for rural entrepreneurs by overcoming infrastructure constraints [10, 12]. This suggests that the differentiated effects observed in Pakistan are likely representative of broader trends across developing contexts.

VI. CONCLUSION

The findings confirm that digitalization—especially mobile access and, to a lesser extent, internet connectivity—significantly promotes entrepreneurship in Pakistan's non-agricultural sector. Mobile access shows a strong and consistent positive effect, while internet access has a modest positive association. In contrast, computer access is negatively related to entrepreneurship, suggesting stronger links to formal employment than to informal enterprise. Heterogeneity analysis indicates that these effects vary by gender and location: mobile and internet access are more empowering for rural residents and men, while women benefit less, reflecting persistent gender-based digital barriers.

The findings of this study carry significant implications for digital inclusion and entrepreneurship development policies in developing economies with similar digital divides—such as Pakistan, Bangladesh, and Kenya—where mobile-first connectivity, affordable broadband, and women-focused digital literacy can translate digital access into entrepreneurial activity [10, 12]. First, the strong positive effect of mobile access on entrepreneurship highlights the need to prioritize mobile-centric policies, especially in rural areas where infrastructure constraints limit access to other technologies. The modest but consistent role of internet connectivity suggests that expanding broadband access and improving affordability could further enhance entrepreneurial opportunities. However, the negative association between computer access and entrepreneurship implies that PC-based interventions may not effectively support informal or small-scale enterprises. Additionally, the gender disparities in the impact of digital tools underscore the importance of gender-responsive strategies, including digital skills training, online safety measures, and tailored support systems to empower women entrepreneurs in the digital economy. These findings emphasize the need for targeted digital inclusion strategies that account for gender disparities and regional inequalities. Policy efforts should prioritize mobile and internet access in underserved communities, alongside digital literacy initiatives, to foster inclusive and equitable entrepreneurial ecosystems.

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